

|   |                                                                                                                                        |    |     |
|---|----------------------------------------------------------------------------------------------------------------------------------------|----|-----|
| 2 | (a) (i) $V_H = 12.4 \cos 36^\circ (= 10.0 \text{ m s}^{-1})$<br>distance = $10.0 \times 0.17$<br>= 1.7 m                               | C1 |     |
|   |                                                                                                                                        | A1 | [2] |
|   | (ii) $V_V = 12.4 \sin 36^\circ (= 7.29 \text{ m s}^{-1})$<br>$h = 7.29 \times 0.17 - \frac{1}{2} \times 9.81 \times 0.17^2$<br>= 1.1 m | C1 |     |
|   |                                                                                                                                        | C1 |     |
|   |                                                                                                                                        | A1 | [3] |
|   | (b) smooth curve with ball hitting wall below original                                                                                 | B1 |     |
|   | smooth curve showing rebound to ground with correct reflection at wall                                                                 | B1 | [2] |
| 3 | (a) point at which (whole) weight (of body) (allow mass for weight)                                                                    | M1 |     |